EE P 538
Additional Guidance for Phase 2

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Overview

- Filter Design in Phase 2:
  - Determine filter requirements given (a) Phase 2 S/N (75dB) requirement, (b) Phase 1 spectral noise output, and (c) Phase 2 passband requirement
  - In your design, remember to EXCEED the min. or max. requirements because components inevitably degrade performance
- Noise Simulations:
  - Use LT Spice for simulating spectral noise density ($V/\sqrt{Hz}$) and total (integrated) noise ($V\ rms$) for frequencies of 0.1Hz to 100MHz.
  - Phase 2 deliverables specify ADC, which LT Spice can not simulate. Thus, Phase 2 instructions included Python code to simulate S/N at ADC output over a smaller frequency range (than LT spice)

Filter Design (1 of 3)

- The max. noise output from the filter ($V_{FiltOut-rms}$ in volts) from Phase 2 S/N:

$$V_{FiltOutMax-rms} = \frac{V_{signal-rms}}{10^{SNR/20}} = \frac{2V/(2\sqrt{2})}{10^{75/20}}$$

- From Phase 1, most students have a design with referred-to-input spectral noise density of $\sim 10nV/\sqrt{Hz}$ or lower. With the voltage gain from Phase 1 of $G_v = 100$, the spectral noise density at the <u>*output*</u> of Phase 1 is:

$$v_{ph1output-rms} \cong 1\mu V/\sqrt{Hz}$$

- Compute the *maximum* acceptable equivalent noise bandwidth of your filter:

$$f_{enb-max} = \left[\frac{V_{FiltOutMax-rms}}{v_{ph1output-rms}}\right]^2$$

Filter Design (2 of 3)

Butterworth (fco = 3 dB)

Order	EqNBW
1	1.5708
2	1.1107
3	1.0472
4	1.0262
5	1.0166
6	1.0115
7	1.0084
8	1.0065
9	1.0051
10	1.0041

Chebyshev (fco = ripple)

Ripple / Order	0.01 dB	0.1 dB	0.25 dB	0.5 dB	1.0 dB
2	3.6672	2.1444	1.7449	1.4889	1.2532
3	1.9642	1.4418	1.2825	1.1666	1.0411
4	1.5039	1.2326	1.1405	1.0656	0.9735
5	1.3114	1.1417	1.0780	1.0208	0.9433
6	1.2120	1.0937	1.0448	0.9970	0.9272
7	1.1537	1.0653	1.0251	0.9828	0.9175
8	1.1166	1.0471	1.0125	0.9736	0.91133
9	1.0914	1.0347	1.0038	0.9674	0.9071
10	1.0736	1.0258	0.9977	0.9629	0.9041

Bessel (fco = 3 dB)

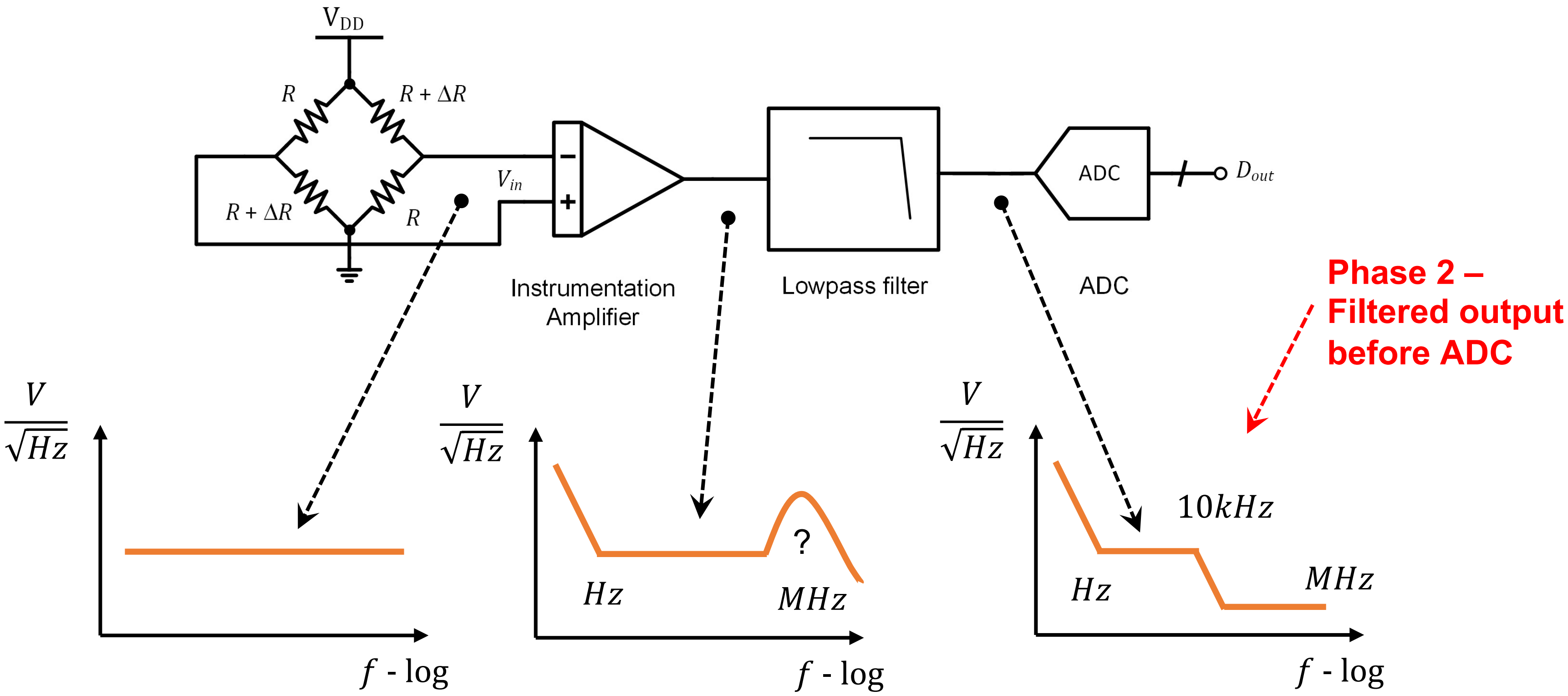
Order	EqNBW
1	1.57
2	1.56
3	1.08
4	1.04
5	1.04
6	1.04

- With maximum effective noise bandwidth ($f_{enb-max}$) and passband requirement ($0.5dB$ at $10KHz$), select the filter type (Bessel, Chebyshev, Butterworth) and the filter order (1,2,4, etc) from a chart or online calculator design tool
- Typically, design tools use a normalize frequency equal to the effective noise bandwidth (f_{enb}) divided by the filter cutoff frequency (f_{co}). *Note this chart (https://www.rfcafe.com/references/electrical/filter-eqnbw.htm) specifies the attenuation (e.g. 3dB or 0.5dB) at the cutoff frequency.*

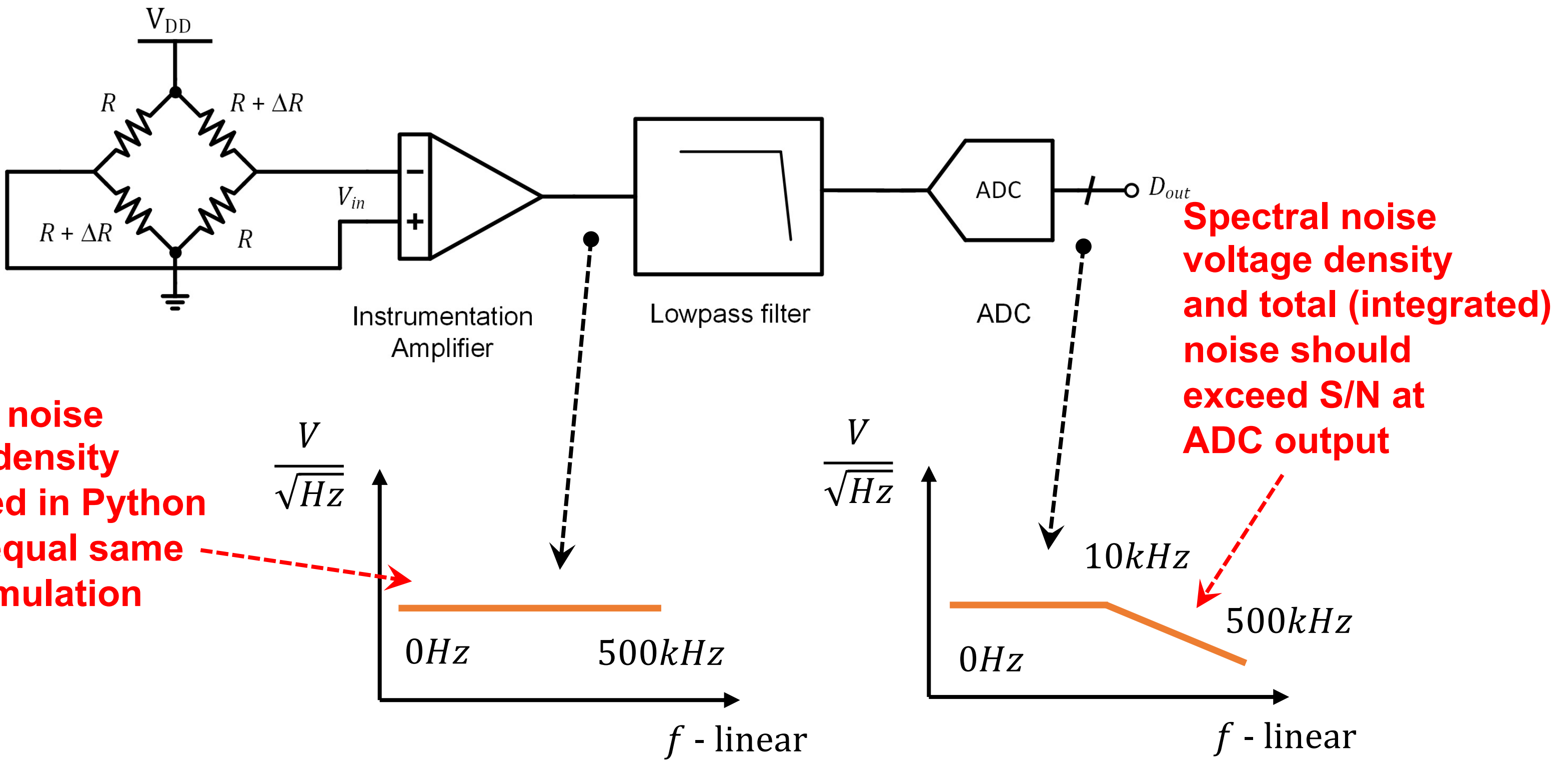
Filter Design (3 of 3)

- In some of your Phase 1 designs, the spectral noise density increases at much higher (e.g. 10MHz) frequencies
- After first-pass implementation of your filter in Phase 2 in LT Spice, observe the noise density again (using LT Spice) at the output of the filter at high frequencies
- If large noise density at high frequencies continue to exist, consider adding in-line low-pass filters using passive devices (resistor & capacitor) at the output of your first-pass implementation.
- These passive filters should have a higher corner frequency (e.g. 50kHz) that will significantly attenuate the 1MHz+ frequencies and will NOT impact the passband of your first-pass filter. (And use LT Spice to check after the second pass implementation.)

Noise Simulation (LT Spice only)



Noise Simulation (Python+Spice) 1 of 3



Spectral noise voltage density generated in Python should equal same as LT simulation

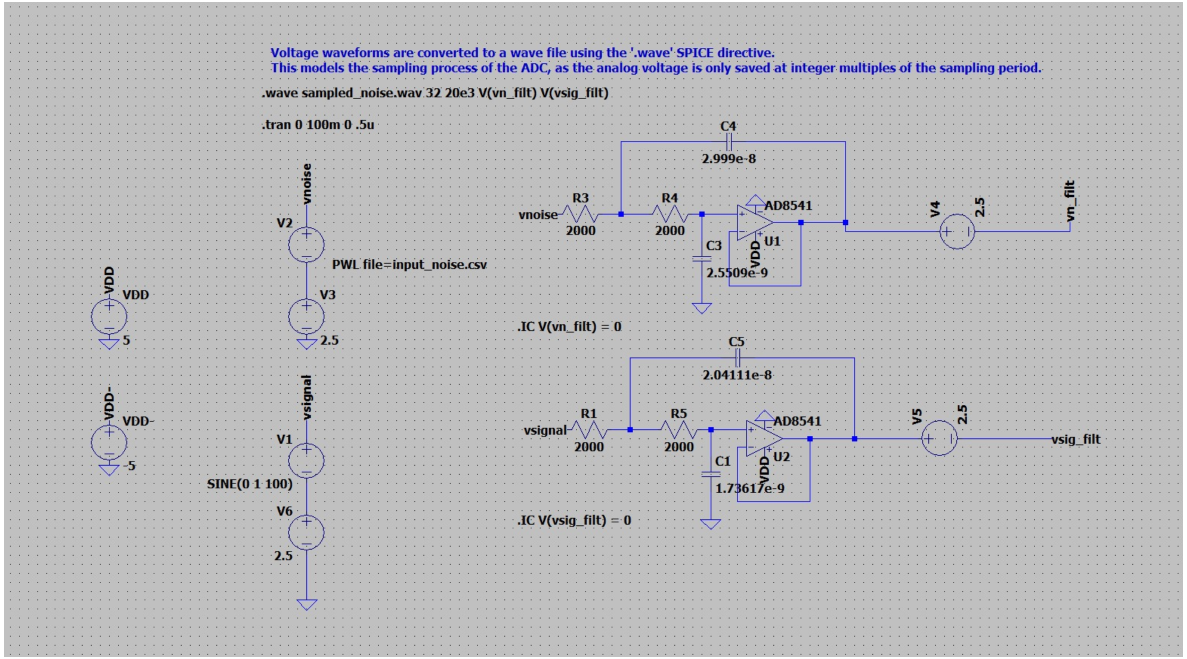
Spectral noise voltage density and total (integrated) noise should exceed S/N at ADC output

Noise Simulation (Python+Spice) 2 of 3

1) Use Python to generate flat spectral noise signal from DC to 500kHz, with a spectral noise amplitude that equals the spectral noise of the Instrumentation Amplifier in the 100Hz-500kHz region. *Tip: Set vn_{rms} in Python equal to the integrated ~100Hz-500kHz noise from output of Instrumentation Amplifier. (And check that spectral noise level makes sense!)*
2) Using output from (1) as the input to the Lowpass Filter, use LT spice to to simulate the output from the Filter. In addition, use LT spice with simulate output of the Filter given an input of a noiseless (desired) signal. *Tip: use ideal voltage supplies to shift voltage up and down in LT Spice as described in next slide*
3) Import the two results from (2) into to Python to determine the S/N after the Python code that 'quantizes' and simulates an ADC

Noise Simulation (Python+Spice) 3 of 3

- *Issue*: Output/input of Python sample uses DC average of zero volts, and WAVE format limits signals to +-1volt
- *Solution*: Use 2.5V ideal voltage sources in Spice to shift voltage level up by 2.5V before filter and shift voltage down by 2.5V after filter



Spectral and total noise voltages

Input/Output stage of Top Level Design	Approximate voltage noise	Noise Bandwidth	Description
Input into Instrumentation Amp (referred-to-input noise)	$\sim 10nV/\sqrt{Hz}$	Not applicable	Approximate maximum spectral noise voltage from a few Hz to hundreds of kHz
Input into Instrumentation Amp (referred-to-input noise)	$\sim 1\mu V$	$0.1Hz - 10kHz$	Using a perfect 10kHz filter, this integrated noise voltage will provide ~77dB of S/N with 20mV p-to-p signal
Input into Instrumentation Amp (referred-to-input noise)	$\sim 7.1\mu V$	$0.1Hz - 500kHz$	The equivalent noise voltage referred to input assuming 500kHz filter
Output from Instrumentation Amp which is Input into Filter	$\sim 1\mu V/\sqrt{Hz}$	Not applicable	Because of gain, note spectral noise voltage is x100 higher than input above
Output from Instrumentation Amp which is Input into Filter	$\sim 0.1mV$	$0.1Hz - 10kHz$	Noise voltage is x100 larger than input above for the same bandwidth
Output from Instrumentation Amp which is Input into Filter	$\sim 0.7mV$	$0.1Hz - 500kHz$	Noise voltage is x100 larger than referred to input noise voltage above for the same 500kHz bandwidth. *This is the noise voltage that should be used with Python+LT Spice (variable vn_rms in sample Python code)*